

Shoulder Pathology Study Guide

IS-NPMSD

Bronwyn Sing, DO

This guide has been created from previous lectures and notes. While it was been read through, there still may be errors. Please email me if you have questions or note inaccuracies. Thank you!

Rotator Cuff Tear

Relevant Anatomy

Rotator cuff is composed of four muscles: supraspinatus, infraspinatus, teres minor, and subscapularis. These muscles form a cover around the head of the humerus and function to rotate the arm and stabilize the humeral head against the glenoid. Most common tendon affected is supraspinatus.

Presentation / Symptoms

Insidious onset, worse with overhead activities. Pain usually localized to lateral (deltoid) aspect of shoulder.

Night pain and difficulty sleeping on the affected side are characteristic. Symptoms include weakness and catching pain when raising the arm overhead (60–120° abduction).

Age <40: less likely Age >40: more risk. Age >65 high risk and sometimes atraumatic, hx of rotator cuff disease increased degenerative tear.

Physical Exam

Inspection: back of shoulder may appear sunken with chronic presentation (infraspinatus atrophy).

Palpation: tenderness over the greater tuberosity.

ROM: active ROM painful past 90° abduction (painful arc); passive ROM near normal.

Strength of SITS with Special test: **Weakness** appreciated with Rotator cuff special tests of SITS mm (Empty Can, Drop Arm, Lift off, Infraspinatus resisted strength)

mild weakness suggests partial tear, full weakness suggests full tear.

Diagnostic Testing

X-rays may show a high-riding humeral head. MRI provides information on muscle status and tear size for surgical planning. Tears can also be visualized on ultrasound, tendon defect.

Management / Treatment

Non-operative: physical therapy, OMM, NSAIDs, avoid overhead activity, corticosteroid injections. Failure of 6 weeks of non-surgical treatment warrants further evaluation. Acute traumatic full tears (with complete loss of supraspinatus strength) require surgical evaluation within 8 weeks.

Impingement Syndrome (Rotator Cuff Tendonitis / Shoulder Bursitis)

Relevant Anatomy

The rotator cuff, primarily the supraspinatus tendon, is pulled repetitively under the coracoacromial arch with overhead movement. The bursa and tendons can become inflamed when the space between the acromion and rotator cuff narrows.

Presentation / Symptoms

Gradual onset of anterior and lateral shoulder pain exacerbated by overhead activity. Night pain and difficulty sleeping on the affected side.

Physical Exam

Tenderness over greater tuberosity and subacromial bursa, crepitus with motion. AROM: Pain worse between 90–120° abduction when lowering the arm. PROM- may be able to do but with pain. Positive Neer's and Hawkins' signs, Empty Can with pain only. Strength is usually **normal**.

Diagnostic Testing

Diagnosis is clinical; imaging unnecessary unless uncertain in setting of trauma, other ddx concerning. X-rays are typically normal. Diagnostic subacromial injection may confirm diagnosis (pain relief with anesthetic).

Management / Treatment

Rest, NSAIDs, stretching program focusing on posterior capsule, subacromial corticosteroids if needed, activity modification to avoid overhead motion. OMM to optimize humeral head and scapular motion.

Calcific Tendonitis

Relevant Anatomy

Characterized by calcium deposits within rotator cuff tendons, often supraspinatus. May relate to endocrine factors (DM, thyroid disease).

Presentation / Symptoms

Rapid onset with high level pain at the top or lateral aspect of the shoulder, radiating toward deltoid insertion. Worse at night and with overhead motion. Pain level is high so patient's can present more urgently.

Physical Exam

During acute pain episodes, patient may hold arm close to body in internal rotation. Positive Neer's and Hawkins' signs.

Diagnostic Testing

X-rays reveal calcifications along shoulder tendons. Clear margins suggest chronic phase; fluffy/poorly defined margins indicate resorptive phase. Ultrasound: calcification of tendon.

Management / Treatment

Similar to impingement treatment: **NSAIDs**, rest, PT, OMM, and possible corticosteroid injection.

Adhesive Capsulitis (Frozen Shoulder)

Relevant Anatomy

Idiopathic loss of both active and passive shoulder motion, still not well understood. Often involves contracture of the coracohumeral ligament. Common ages 40–60, more frequent in women. Associated with diabetes, thyroid disease, prolonged immobilization.

Presentation / Symptoms

Severe, nagging night pain, progressive stiffness, and marked restriction of activities of daily living.

Stages: Freezing (painful),

Frozen (stiffness with less pain),

Thawing (gradual improvement). Course lasts 6–24 months!

Physical Exam

At least 50% reduction in both active and passive ROM. **Loss of external rotation** with arm at side is pathognomonic. Loss of arm swing with gait. Pain and tenderness at deltoid insertion.

Diagnostic Testing

Diagnosis is clinical. But AP and axillary radiographs to ensure smooth joint surfaces and exclude glenohumeral OA, loose bodies, or calcifications, masses.

Management / Treatment

NSAIDs, non-narcotic analgesics, moist heat followed by gentle stretching, TENS, PT/home stretching 3–6 months, corticosteroid injections. Non-surgical treatment successful in 80–85%. OMM and PT for mobility restoration.

Acromioclavicular (AC) Joint Separation

Relevant Anatomy

AC joint connects the clavicle and acromion. Stabilized by acromioclavicular and coracoclavicular ligaments.

Presentation / Symptoms

Results from a fall onto the tip of the shoulder. Pain over AC joint, pain with lifting arm. Deformity may be present (Types III–VI).

Physical Exam

Pain with palpation over AC joint and with traction on a hanging arm. Pain with lifting arm.

Diagnostic Testing

AP X-rays of both shoulders confirm separation and classify by degree of displacement (Types I–VI).

Management / Treatment

Type I–II: conservative (sling, ice, analgesics, return to activity in ~4 weeks). Type III: surgical vs non-surgical based on function and cosmetic concern. Type IV–VI: surgical repair.

Shoulder Instability (Anterior and Posterior Dislocation)

Relevant Anatomy

Shoulder instability may be anterior, posterior, inferior, or multidirectional. Anterior dislocation occurs with forceful abduction and external rotation (sports, falls).

Presentation / Symptoms

Acute anterior dislocation: trauma, instability, and pain. Arm in abduction gets pulled backwards and anterior portion of shoulder pulls forward.

Posterior dislocation: pain with flexion/adduction/internal rotation (often after seizure or shock). Once there has been a dislocation, there will always be laxity. Recurrent instability presents as sense of shoulder slipping out of joint, patient's will guard when they sense it is going to happen.

Physical Exam

Anterior: positive apprehension test. Posterior: limited external rotation, arm held internally rotated. Multidirectional: sulcus sign positive.

Diagnostic Testing

X-rays (AP, scapular Y, axillary views). MRI if labral involvement suspected. Associated lesions: **Bankart (labral avulsion)**, **Hill-Sachs (humeral impaction)**.

Management / Treatment

Reduction in ED, immobilization 3 weeks in neutral, followed by PT. Surgical evaluation for recurrent or complex instability.

Labral Injuries (SLAP Lesions)

Relevant Anatomy

Superior Labrum Anterior-to-Posterior (SLAP) lesions involve injury to superior glenoid labrum and biceps anchor complex.

Presentation / Symptoms

Vague deep shoulder pain, painful popping or catching, weakness or fatigue with use. Often seen in throwing athletes or after fall on outstretched arm.

Physical Exam

Positive O'Brien's, Crank, Clunk, or Resisted Supination/ER tests (low sensitivity/specificity).

Diagnostic Testing

Plain radiographs are initial but non-diagnostic. MR arthrography is gold standard (90% sensitivity/specificity). Definitive diagnosis via arthroscopy.

Management / Treatment

Non-surgical first: NSAIDs, stretching, strengthening, and gradual return to activity. Arthroscopy if persistent symptoms.

Biceps Tendon Rupture

Relevant Anatomy

Rupture of the proximal long head of biceps, often associated with chronic rotator cuff disease.

Presentation / Symptoms

Sudden upper arm pain with audible snap. Bulge in lower arm (Popeye sign). Pain mild after acute phase.

Physical Exam

Visible bulge accentuated with resisted flexion. Palpable defect proximally. Ecchymosis tracking down arm (acute).

Diagnostic Testing

AP and axillary radiographs to exclude fracture. MRI to rule out concomitant rotator cuff tear.

Management / Treatment

Non-surgical for most (little functional loss). Surgical repair for athletes or adults <40 years old.

Biceps Tendonitis

Relevant Anatomy

Inflammation of proximal or distal biceps tendon, often with subacromial bursitis.

Presentation / Symptoms

Anterior shoulder pain worsened with lifting or repetitive flexion/supination.

Physical Exam

Tenderness over bicipital groove. Positive Speed's and Yergason's tests.

Diagnostic Testing

Diagnosis is clinical. Imaging only if other pathology suspected.

Management / Treatment

Ice, heat, stretching, physical therapy, and corticosteroid injection for persistent cases.

Accessory Muscle Strain/Spasm

Relevant Anatomy

Common in trapezius or rhomboids, often secondary to lifting, cervical OA, or torticollis.

Presentation / Symptoms

Localized pain and stiffness in upper back/neck region.

Physical Exam

Tenderness and muscle tightness on palpation.

Diagnostic Testing

Diagnosis is clinical.

Management / Treatment

Ice, heat, gentle stretching, OMM, hydration, NSAIDs, and muscle relaxers as needed.